object stores offer a simple HTTP API to store and list objects. Different cloud providers offer a range of signals from object stores, usually at least logs and metrics—for example, Amazon S3 (http://mng.bz/Y1Xz), Azure Blob Storage (http://mng.bz/GylD), and Google Cloud Storage (https://cloud.google.com/storage/docs/access-logs).

We've now completed a brief review of storage-related sources, which is mostly relevant for infrastructure operators and is secondary for developers. We now move on to a kind of signal source that you will have to pay attention to, no matter what your role is: networks and their signals.

3.4 Network-related sources

In this section, we will look at all sorts of network-related sources, from low-level interfaces to virtual networks. As with storage, these sources are mostly owned by others (e.g., your cloud provider of choice) and, by their nature, are high-volume sources, meaning you have to deal with a lot of noise.

3.4.1 Network interfaces

Physical network interfaces or virtual interfaces, such as the loopback interface, form the lowest layer of network-related sources. Depending on the environment (think serverless data planes or FaaS), you may or may not have direct access to these interfaces. By *direct access*, I mean being able to use commands such as `ip` (http://mng.bz/zXo1) to query interface and routing information.

To demonstrate the difference between having direct access to the network interface information and when that's not the case, imagine you have an EC2 instance (e.g., as part of a Kubernetes cluster data plane) that you can `ssh` into and then execute (the output is edited to fit):

Using the ip command, list network interfaces, their configuration, and status.

```
$ ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode
    DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP mode
    DEFAULT group default qlen 1000
    link/ether 0a:78:c2:54:1f:39 brd ff:ff:ff:ff:ff:ff
    altname enp0s5
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state
    DOWN mode DEFAULT group default
    link/ether 02:42:18:7d:57:00 brd ff:ff:ff:ff:ff:ff
```

```
$ ip route
default via 172.31.32.1 dev ens5 proto dhcp src 172.31.40.108 metric 100
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown
172.31.32.0/20 dev ens5 proto kernel scope link src 172.31.40.108
172.31.32.1 dev ens5 proto dhcp scope link src 172.31.40.108 metric 100
```

Using the ip command, list routing information, such as the default gateway.